7	**(a)**	A photon is defined as a <u>quantum of electromagnetic energy.</u>	
	(b)	**(i)**	1.0 V By C.O.E, EPE = KE_max EPE = (1.0)(1.6×10^{-19}) = 1.6×10^{-19} J
		(ii)	From principle of conservation of energy, hf = Φ + KEmax KEmax = eVs = ½ mv^2 [M1] v = $\sqrt{\frac{(2(1.6\times10-19)(1))}{(9.11\times10-31)}}$ = 592673 ~ 590 km s^{-1} [A1]
		(iii)	For Φ, E = hc/λ = $\frac{(6.63\times10-34)(3.00\times108)}{365\times10-9}$=5.4493$\times 10-19$ [M1] = 3.4058 ~ 3.4 eV Φ = 3.4058 – 1.0 eV ~ 2.4 eV [A1]